



GTC4Lusso USA - Engine coolant pump - Latest Update: 2021-01-26

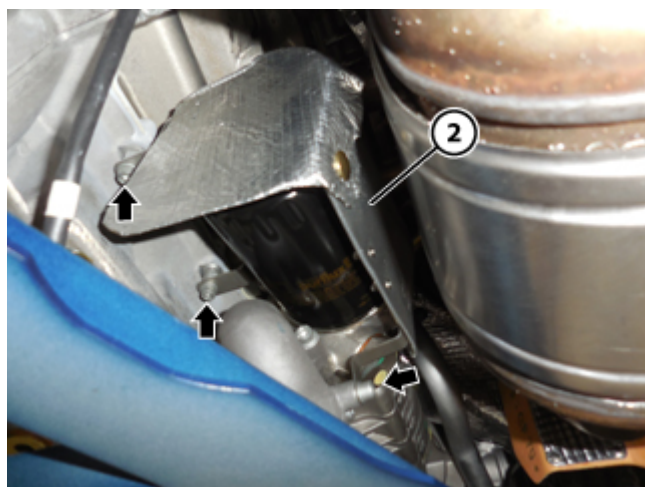
Protected by copyright. Copying for private or commercial purposes, in part or in whole, is not permitted unless authorised by Ferrari S.p.A. Ferrari S.p.A. does not guarantee or accept any liability with respect to the correctness of information in this document.

Copyright by Ferrari S.p.A. - P.IVA 00159560366

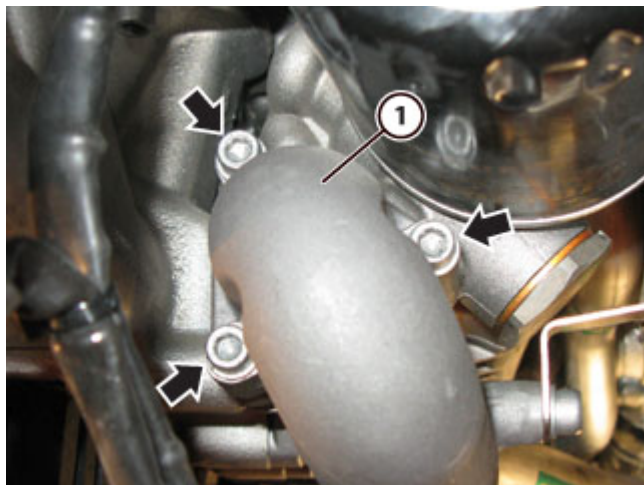
B6.03 Engine coolant pump

Removing the water/oil pump

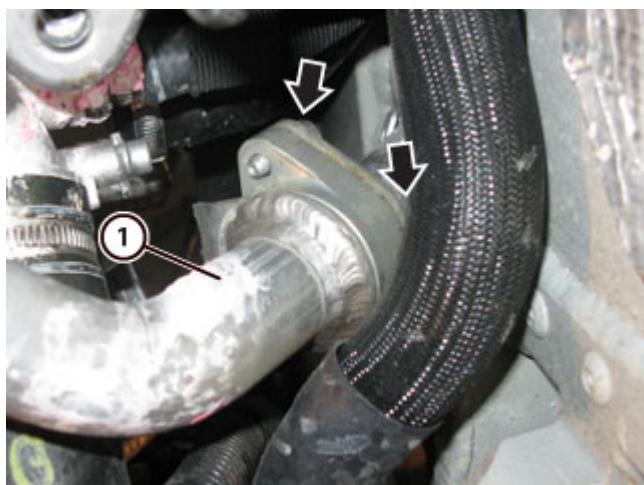
- Remove the front wheelhouses (🔑 E3.05).
 - ❗ Remove the front right hand wheelhouse.
- Drain the engine cooling system (🔑 B6.07).
- Drain the engine lubrication system (🔑 A3.01).
- Remove the front removable subframe (🔑 E2.02).
- Remove the catalytic converters (🔑 B7.03).
 - ❗ Remove the right hand catalytic converter.



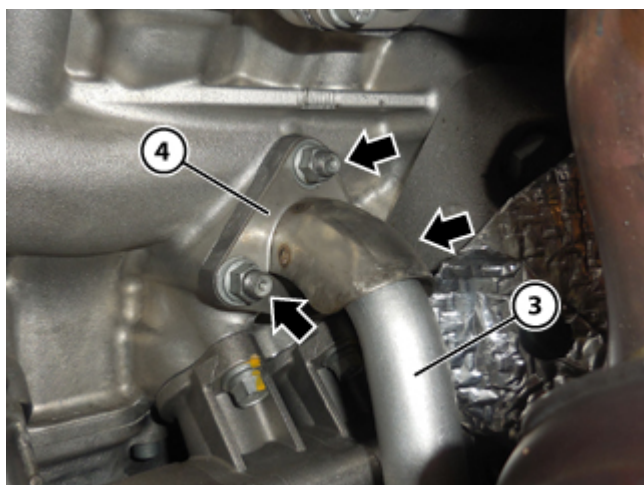
- Remove the heat shield (2), by undoing the screws indicated.



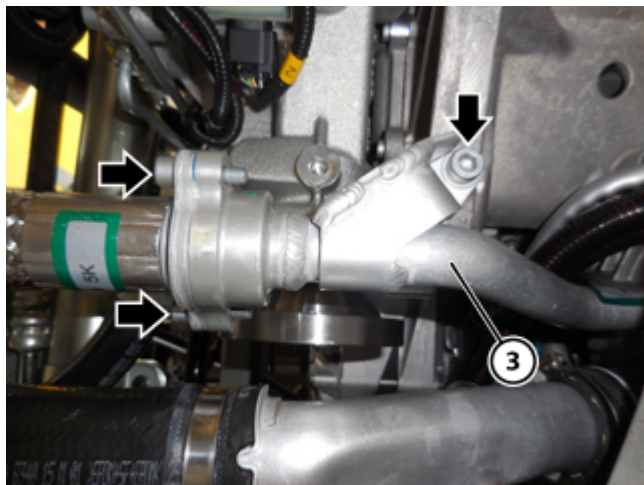
- Undo the indicated nuts on the engine oil pipe (1).



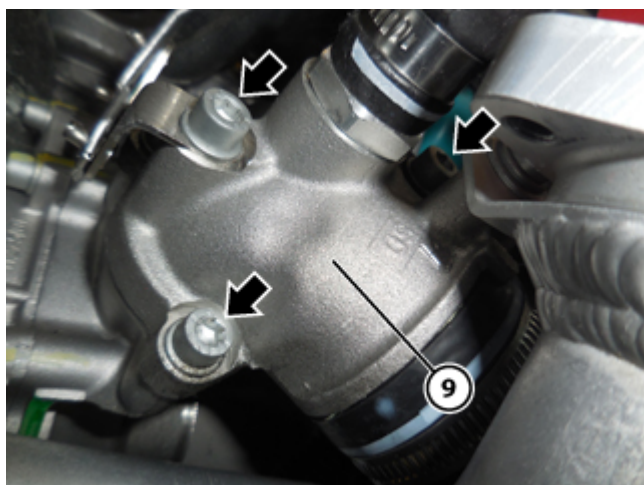
- Undo the indicated screws and remove the engine oil pipe (1).



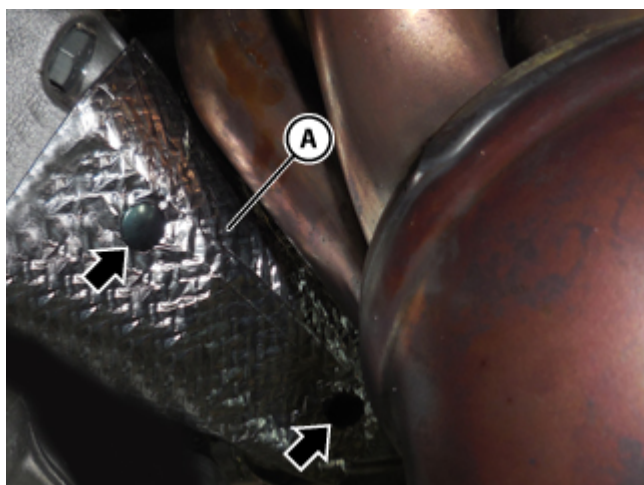
- Undo the indicated nuts fastening the engine oil pipe (3) to the crankcase and retrieve the shield (4).



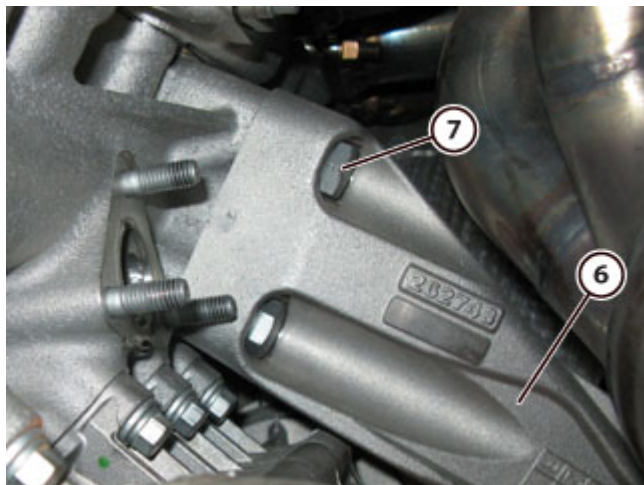
- Undo the indicated screw.
- Remove the engine oil pipe (3), undoing the indicated screws.



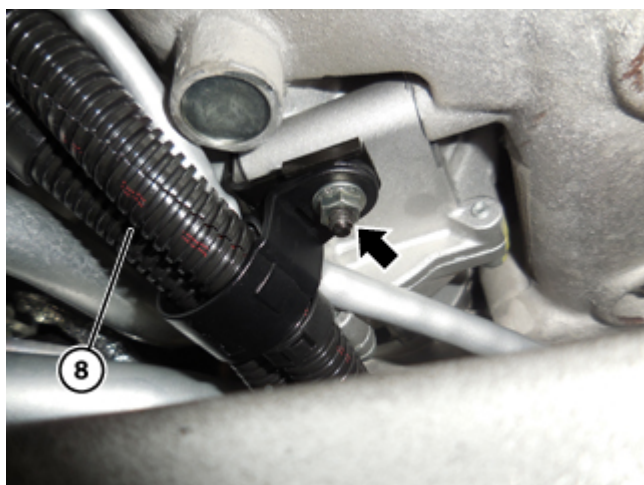
- Remove the hose union (9) from the pump, undoing the indicated screws.



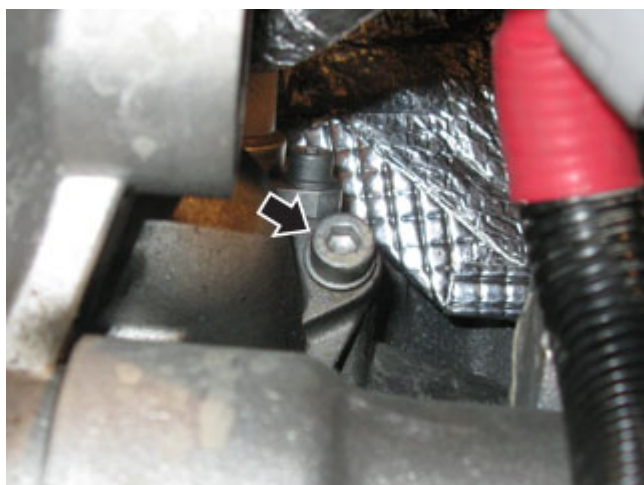
- Remove the heat shield (A), releasing the indicated clips.



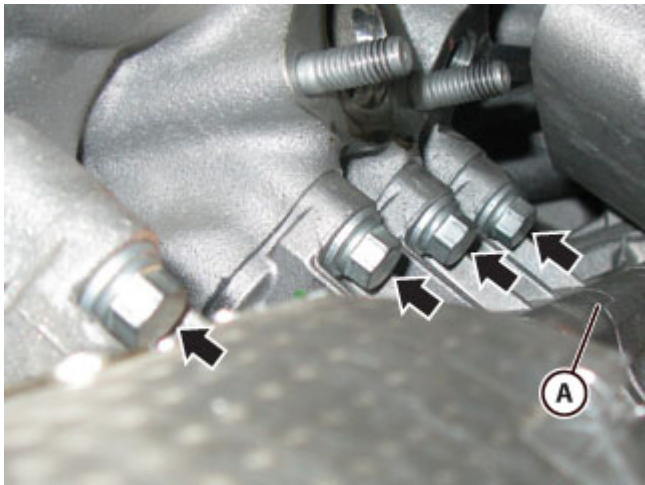
- Undo the four screws (7) and move the engine mounting strut (6) aside.



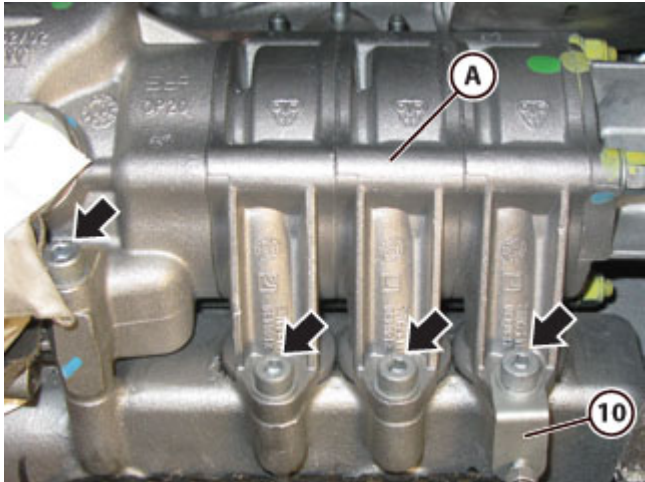
- Undo the indicated screw to release the engine services cable (8).



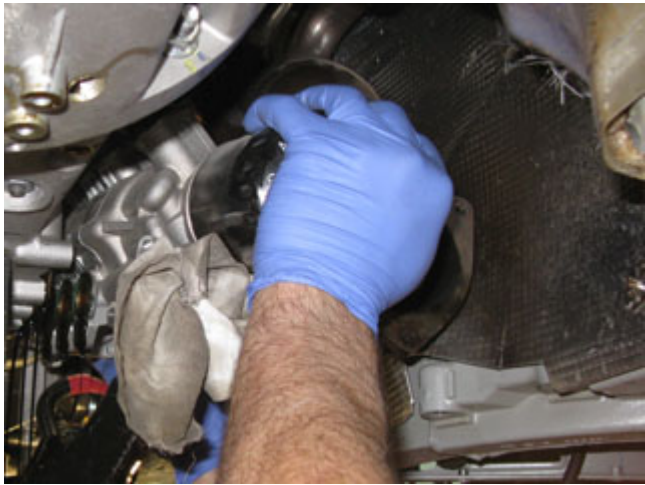
- Undo the indicated screw.



- Undo the indicated screws fastening the water/oil pump (A).



- Undo the indicated screws fastening the water/oil pump (A).
- Release the bracket (10).

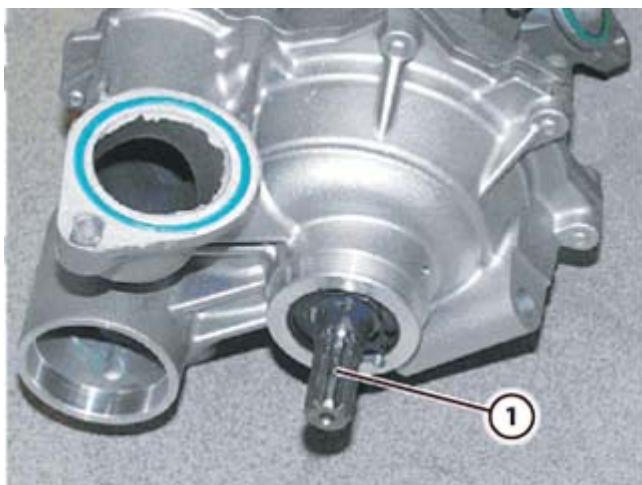


- Ease out the water/oil pump horizontally towards the rear and remove.

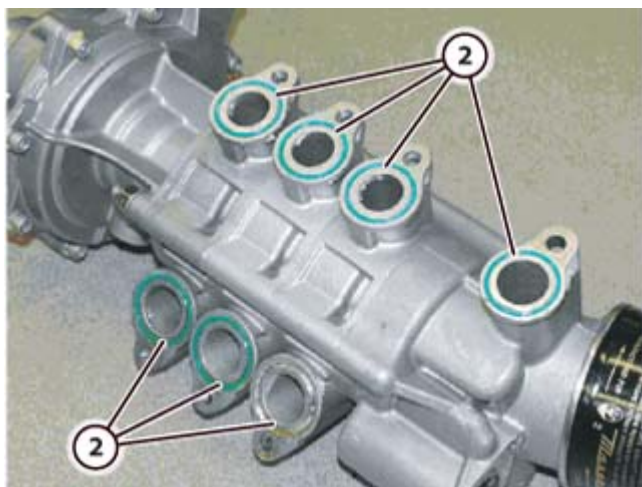
Refitting the water/oil pump

			
Tightening torque		Nm	Class
Fastening water/oil pump to crankcase	Screw (pretightening)	5 Nm	0
	Screw	25 Nm	B

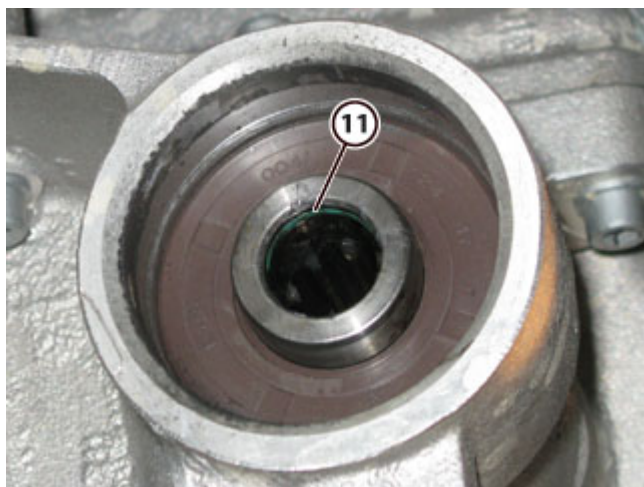
			
Tightening torque		Nm	Class
Fastening engine mounting strut to crankcase	Screw	45 Nm	A
Fastening hose connector to water/oil pump	Screw	10 Nm	B
Fastening oil pipe leading from engine to disconnecter	Nut	20 Nm	B
Fastening engine oil delivery pipe for thermostat to section	Screw	9 Nm	B
Fastening engine oil pipes leading from tank	Screw	10 Nm	B
Fastening engine oil pipe to water/oil pump	Screw	10 Nm	B



- With the water/oil pump on the workbench, check that there are no leaks from the drain orifices.
- When refitting the same oil/water pump, check the state of wear of the splines on the shaft (1).
- Apply a thin film of **MOLYKOTE BR2** on the shaft (1).

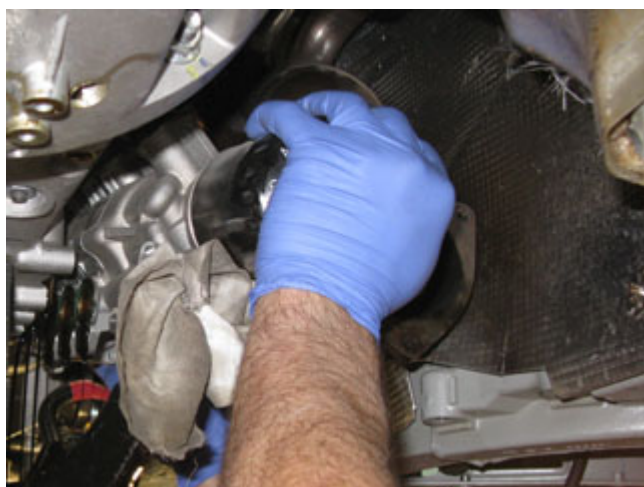


- Apply a thin film of **SHELL ALVANIA RL3** to the O-ring seats before installing the O-rings.
-  Always replace the O-rings with new components.
- Fit the O-rings (2) on the water/oil pump inlets.



Always replace the O-rings with new components.

- Replace the O-ring (11) on the lower crankcase shell joint and grease the relative seat with **SHELL ALVANIA RL3** before reassembly.



When installing the water/oil pump, take particular care not to pinch the new O-rings and that they remain in their relative seats.

- Fit the complete water/oil pump on the crankcase, inserting the splined shaft into the pump drive shaft.

- Hand-tighten all the indicated screws fastening the water/oil pump.
- The screws must be tightened according to the following procedure and sequence:

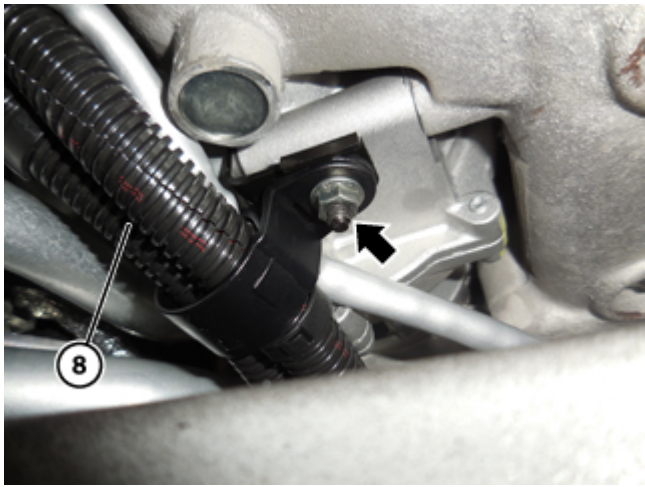
 Pre-tighten all the screws in the sequence given.

 Tighten the screws in the same sequence.



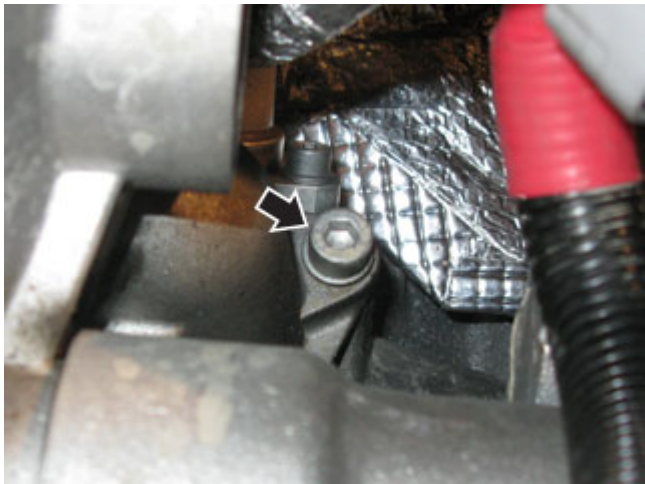
- Tighten the indicated screw.

		
Tightening torque	Nm	Class
Screw (pretightening)	5 Nm	0
Screw	25 Nm	B



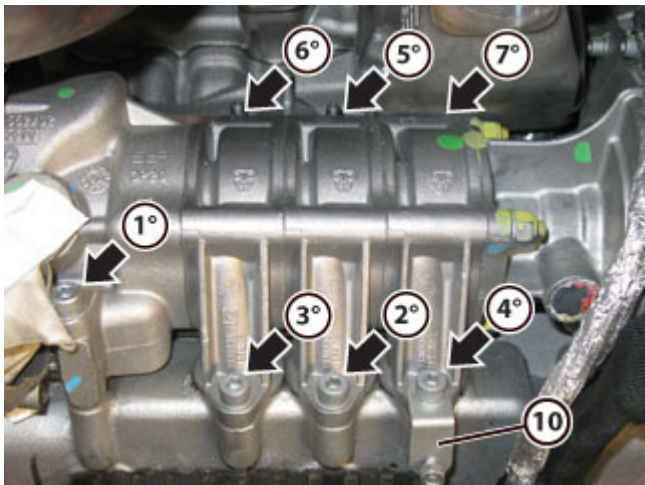
- Tighten the indicated screw fastening the engine services cable (8).

		
Tightening torque	Nm	Class
Screw (pretightening)	5 Nm	0
Screw	25 Nm	B



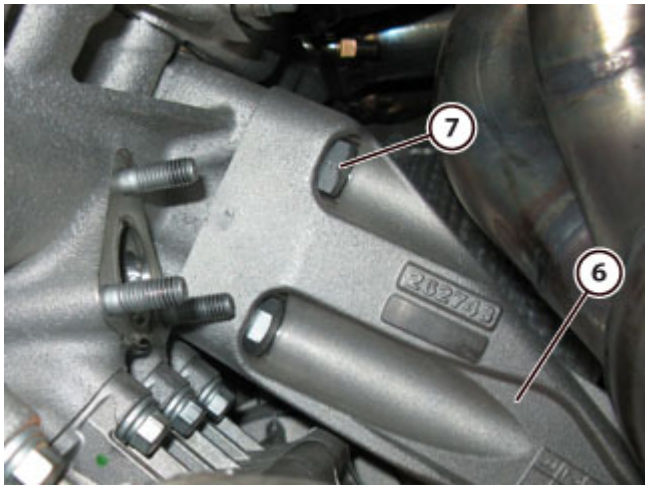
- Tighten the indicated screw.

		
Tightening torque	Nm	Class
Screw (pretightening)	5 Nm	0
Screw	25 Nm	B



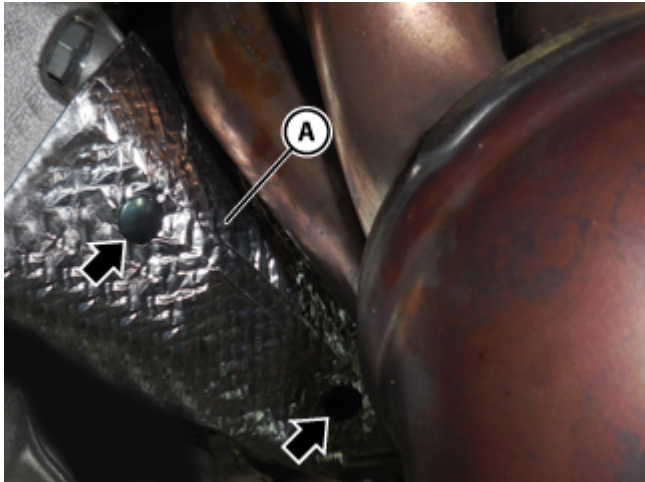
- Fit the bracket (10) then tighten the indicated screws in the sequence given.

		
Tightening torque	Nm	Class
Screw (pretightening)	5 Nm	0
Screw	25 Nm	B

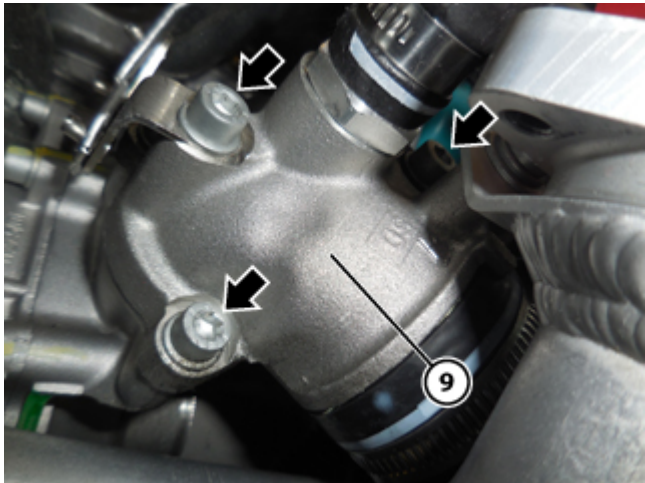


- Fit the engine mounting strut **(6)** in the relative seat and fasten by tightening the four screws **(7)**.

		
Tightening torque	Nm	Class
Screw	45 Nm	A

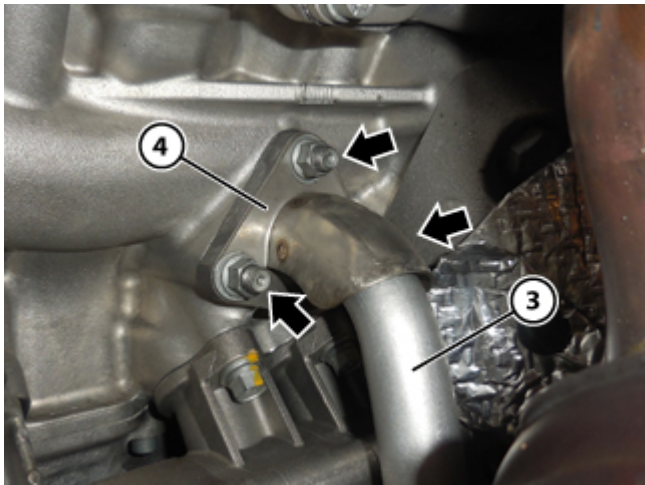


- Fit and fasten the heat shield **(A)**, fastening the indicated clips.



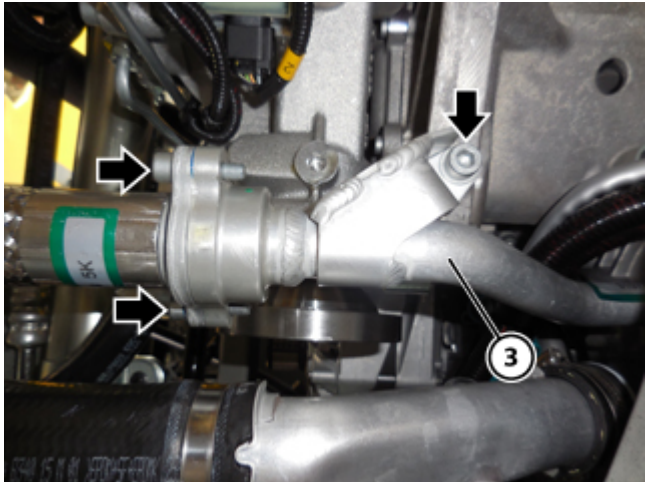
- Apply **SEGO** to the O-ring seat and replace the O-ring.
- Fit the hose union **(9)** on the pump and tighten the indicated screws.

		
Tightening torque	Nm	Class
Screw	10 Nm	B



- Apply **SEGO** to the O-ring seat and replace the O-ring.
- Fit the engine oil pipe **(3)** on the relative studs.
- Fit the shield **(4)** and tighten the indicated nuts.

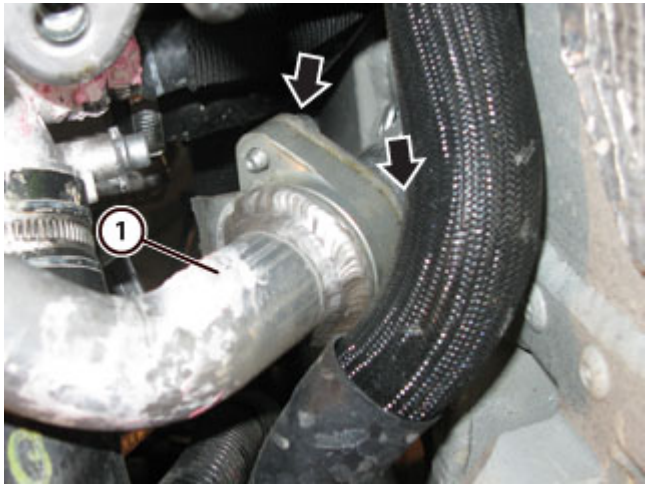
		
Tightening torque	Nm	Class
Nut	20 Nm	B



- Tighten the screws, after applying **AREXONS SYSTEM 52A43 - MEDIUM STRENGTH THREAD LOCK** to the threads.

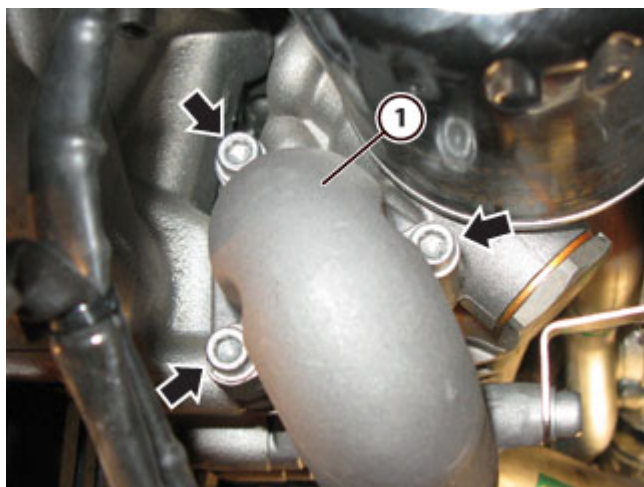
		
Tightening torque	Nm	Class
Screw	9 Nm	B

- Fasten the engine oil pipe with the indicated screw.



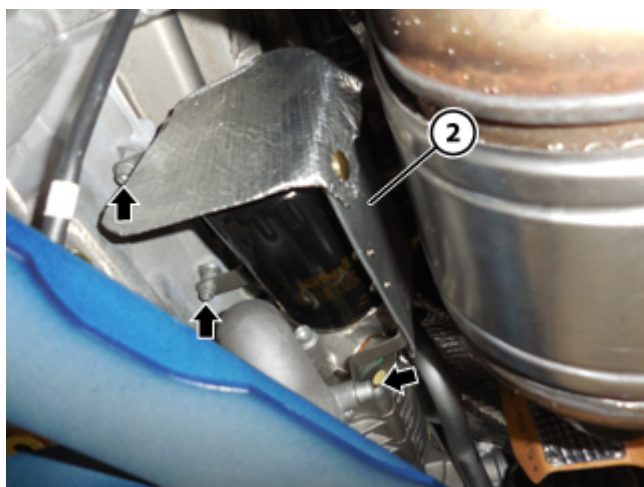
- Apply **SEGO** to the seat on the pipe, replace the O-ring, then connect the engine oil pipe **(1)**, tightening the indicated screws.

		
Tightening torque	Nm	Class
Screw	10 Nm	B



- Apply **SEGO** to the seat on the pump, replace the O-ring, then fasten the engine oil pipe **(1)**, tightening the indicated screws.

		
Tightening torque	Nm	Class
Screw	10 Nm	B



- Fasten the heat shield **(2)** by tightening the indicated screws.

- Refit the catalytic converters ( **B7.03**).
- Refit the right hand catalytic converter.
- Refit the front removable subframe ( **E2.02**).
- Fill the engine lubrication system ( **A3.01**).
- Fill the engine coolant system ( **A3.07**).
- Refit the front wheelhouses ( **E3.05**).
- Refit the front right hand wheelhouse.